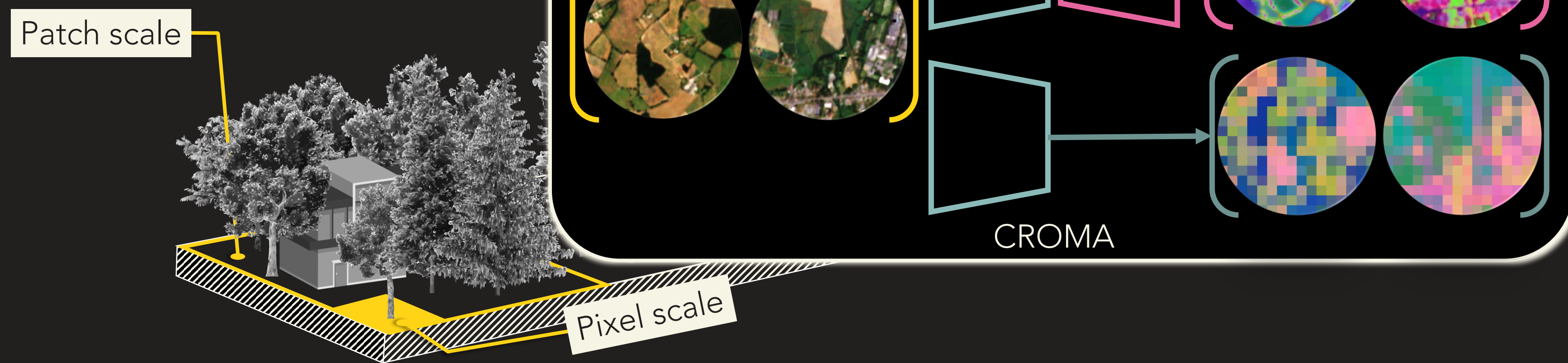


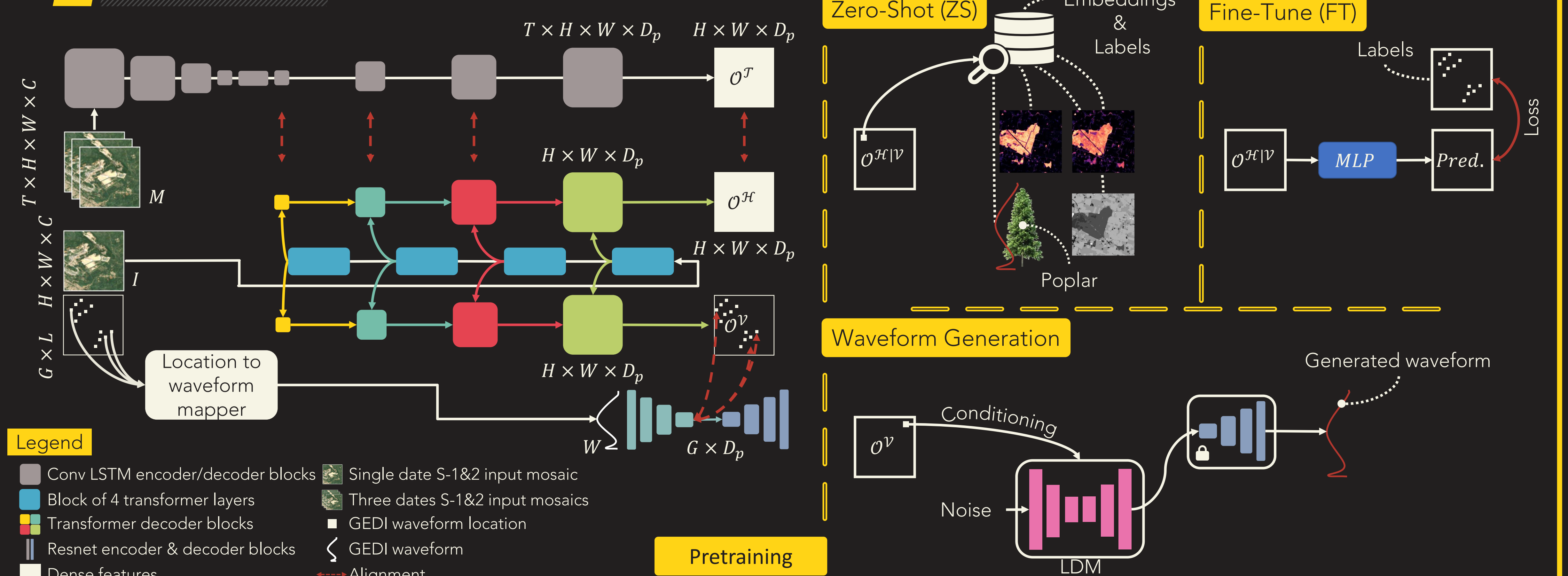


1. Motivation

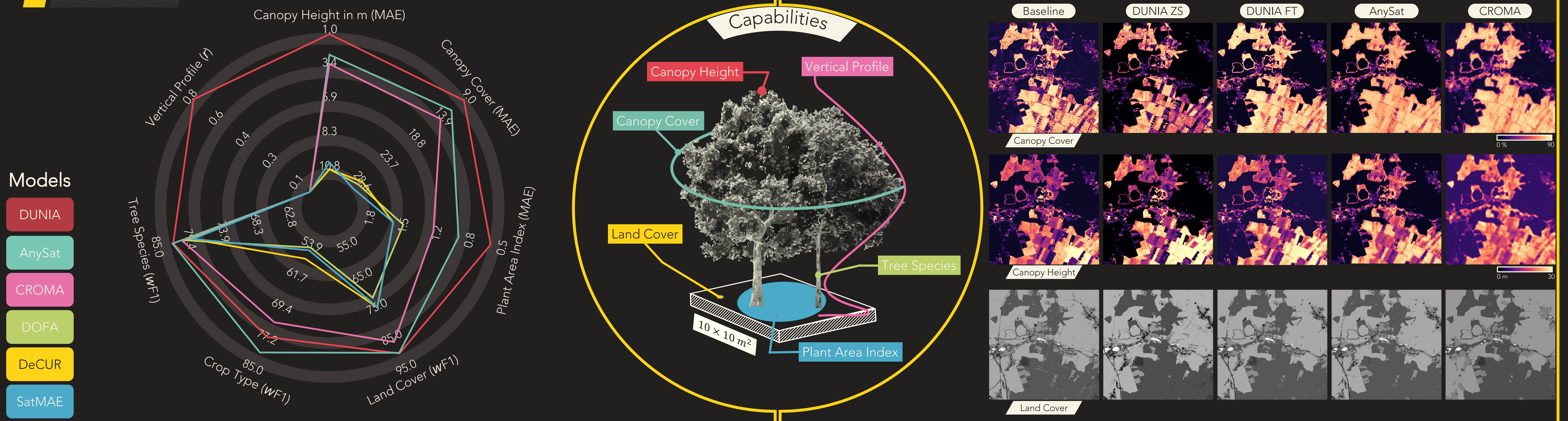
- Labels which are a fundamental part for training any ML model, are few and far between for EO applications.
- Several self-supervised earth observation models currently exist, however:
 - The majority operates at the patch scale, which results in embeddings that are not readily useable.
 - They do not/cannot leverage LiDAR data (e.g., GEDI), essential for applications requiring vertical structure understanding.
- No existing model is able to estimate the full vertical Structure of canopies.



2. Method



3. Results



4. Conclusions

- Contrasting at the pixel level enables zero-shot capabilities that in many cases outperform specialist models.
- By integrating LiDAR data, we recover the full vertical vegetation profile; our approach is the first to achieve complete canopy vertical profiling in a self-supervised framework.
- Dual vertical and horizontal alignment equips the model for tasks demanding both spectral and structural understanding (e.g., biomass estimation, species delineation).
- Compositing cuts out the need for time-series storage, lightening data burdens, though certain applications (e.g., crop type mapping) would greatly benefit from temporal dynamics.
- The architecture is computationally efficient and can be trained with limited compute.

5. References

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6. Acknowledgements

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